

Problem 4

Part a)

To show that $\mathbb{Q}$ is a probability measure on $(\Omega, \mathcal{F})$, we verify the properties from Definition 1.1.9:

1. **Non-negativity:** We have that:

$$\mathbb{Q}(A) = \mathbb{E}_{\mathbb{P}}(X \mathbf{1}_A) = \int_{\Omega} X \mathbf{1}_A d\mathbb{P}$$

Since $X \geq 0$ and $\mathbf{1}_A \geq 0$, then $X \mathbf{1}_A \geq 0$, and, by monotonicity of the integral, $\int_{\Omega} X \mathbf{1}_A d\mathbb{P} \geq 0$. Therefore, $\mathbb{Q}(A) \geq 0, \forall A \in \mathcal{F}$.

2. **Normalization:** For the sample space Ω , $\mathbf{1}_{\Omega} = 1$. Therefore:

$$\mathbb{Q}(\Omega) = \mathbb{E}_{\mathbb{P}}(X \mathbf{1}_{\Omega}) = \mathbb{E}_{\mathbb{P}}(X) = 1$$

3. **σ -additivity:** Let $(A_n)_{n \in \mathbb{N}} \subset \mathcal{F}$ be a sequence of pairwise disjoint events, and let $A = \bigcup_{n=1}^{\infty} A_n$. Then:

$$\mathbb{Q}(A) = \mathbb{E}_{\mathbb{P}}(X \mathbf{1}_A) = \mathbb{E}_{\mathbb{P}}\left(X \mathbf{1}_{\bigcup_{n=1}^{\infty} A_n}\right)$$

Since the sets A_n are pairwise disjoint, then we have that $\mathbf{1}_{\bigcup_{n=1}^{\infty} A_n} = \sum_{n=1}^{\infty} \mathbf{1}_{A_n}$. Then:

$$\mathbb{Q}(A) = \mathbb{E}_{\mathbb{P}}\left(X \sum_{n=1}^{\infty} \mathbf{1}_{A_n}\right) = \mathbb{E}_{\mathbb{P}}\left(\sum_{n=1}^{\infty} X \mathbf{1}_{A_n}\right) = \mathbb{E}_{\mathbb{P}}\left(\lim_{N \rightarrow \infty} \sum_{n=1}^N X \mathbf{1}_{A_n}\right)$$

Since $X \mathbf{1}_{A_n} \geq 0$ for all $n \in \mathbb{N}$, the sequence of partial sums $\left(\sum_{n=1}^N X \mathbf{1}_{A_n}\right)_{N \in \mathbb{N}}$ is monotonically increasing and converges pointwise to $X \mathbf{1}_A$, i.e., $\left(\sum_{n=1}^N X \mathbf{1}_{A_n}\right) \uparrow X \mathbf{1}_A$. By the Monotone Convergence Theorem (Theorem 3.3.11a):

$$\mathbb{Q}(A) = \mathbb{E}_{\mathbb{P}}\left(\lim_{N \rightarrow \infty} \sum_{n=1}^N X \mathbf{1}_{A_n}\right) = \lim_{N \rightarrow \infty} \mathbb{E}_{\mathbb{P}}\left(\sum_{n=1}^N X \mathbf{1}_{A_n}\right)$$

By the linearity of the expected value (Lemma 3.3.3):

$$\lim_{N \rightarrow \infty} \mathbb{E}_{\mathbb{P}}\left(\sum_{n=1}^N X \mathbf{1}_{A_n}\right) = \lim_{N \rightarrow \infty} \sum_{n=1}^N \mathbb{E}_{\mathbb{P}}(X \mathbf{1}_{A_n}) = \sum_{n=1}^{\infty} \mathbb{Q}(A_n)$$

Thus, $\mathbb{Q}$ is a probability measure on $(\Omega, \mathcal{F})$. □

Part b)

Let $A \in \mathcal{F}$ such that $\mathbb{P}(A) = 0$. Then, the indicator function $\mathbf{1}_A$ is equal to 0 with probability 1, i.e., $\mathbf{1}_A = 0$ $\mathbb{P}$ -almost surely ($\mathbb{P}$ -a.s.). Since $X \geq 0$ is a real-valued random variable, the product $X \mathbf{1}_A$ is also a random variable, and, also, $X \mathbf{1}_A = 0$ $\mathbb{P}$ -a.s. By the definition of the expected value (Definition 3.3.8), random variables that are $\mathbb{P}$ -almost surely equivalent possess the same expected value. Hence:

$$\mathbb{Q}(A) = \mathbb{E}_{\mathbb{P}}(X \mathbf{1}_A) = \mathbb{E}_{\mathbb{P}}(0) = 0. \quad \square$$

Counterexample Let $\Omega = \{1, 2\}$ with the σ -algebra $\mathcal{F} = \mathcal{P}(\Omega)$ and let the probability measure $\mathbb{P}$ be:

$$\mathbb{P}(\{1\}) = 0.5, \quad \mathbb{P}(\{2\}) = 0.5.$$

Define the random variable $X : \Omega \rightarrow \mathbb{R}$ by:

$$X(1) = 2, \quad X(2) = 0.$$

We have that $X \geq 0$, and we verify the normalization condition:

$$\mathbb{E}_{\mathbb{P}}(X) = \sum_{\omega \in \{1, 2\}} X(\omega) \mathbb{P}(\{\omega\}) = 2(0.5) + 0(0.5) = 1.$$

Now, we define the event $A = \{2\} \in \mathcal{F}$. The probability of A under $\mathbb{P}$ is strictly positive:

$$\mathbb{P}(A) = \mathbb{P}(\{2\}) = 0.5 > 0.$$

We now compute $\mathbb{Q}(A)$ using the definition of the expected value for a discrete simple random variable:

$$\begin{aligned} \mathbb{Q}(A) &= \mathbb{E}_{\mathbb{P}}(X \mathbf{1}_A) \\ &= \sum_{\omega \in \{1, 2\}} X(\omega) \mathbf{1}_A(\omega) \mathbb{P}(\{\omega\}) \\ &= X(1) \mathbf{1}_{\{2\}}(1) \mathbb{P}(\{1\}) + X(2) \mathbf{1}_{\{2\}}(2) \mathbb{P}(\{2\}) \\ &= (2 \cdot 0 \cdot 0.5) + (0 \cdot 1 \cdot 0.5) \\ &= 0 + 0 = 0. \end{aligned}$$

Thus, we have found an event A such that $\mathbb{P}(A) > 0$ but $\mathbb{Q}(A) = 0$, which shows that the converse does not hold.

Part c)

Using measure-theoretic induction:

- **Step 1. Indicator functions:** Let $Y = \mathbf{1}_A$, for $A \in \mathcal{F}$. By definition of $\mathbb{Q}$ and the expected value of an indicator function (Remark 3.3.2):

$$\mathbb{E}_{\mathbb{Q}}(\mathbf{1}_A) = \mathbb{Q}(A) = \mathbb{E}_{\mathbb{P}}(X \mathbf{1}_A)$$

- **Step 2. Simple functions:** Let $Y = \sum_{i=1}^n a_i \mathbf{1}_{A_i}$ be a simple, non-negative random variable ($a_i \geq 0$). By the linearity of the expected value (Lemma 3.3.3):

$$\mathbb{E}_{\mathbb{Q}}(Y) \stackrel{(Lin.)}{=} \sum_{i=1}^n a_i \mathbb{E}_{\mathbb{Q}}(\mathbf{1}_{A_i}) \stackrel{(\text{Step 1})}{=} \sum_{i=1}^n a_i \mathbb{E}_{\mathbb{P}}(X \mathbf{1}_{A_i}) \stackrel{(Lin.)}{=} \mathbb{E}_{\mathbb{P}}\left(X \sum_{i=1}^n a_i \mathbf{1}_{A_i}\right) = \mathbb{E}_{\mathbb{P}}(XY)$$

- **Step 3. Non-negative random variables:** Let $Y \geq 0$ be an arbitrary non-negative random variable. By Lemma 3.3.4, there exists a monotonically increasing sequence of simple non-negative random variables $(Y_n)_{n \in \mathbb{N}}$ such that $Y_n \uparrow Y$. By the Monotone Convergence Theorem (Theorem 3.3.11a) applied under $\mathbb{Q}$:

$$\mathbb{E}_{\mathbb{Q}}(Y) = \lim_{n \rightarrow \infty} \mathbb{E}_{\mathbb{Q}}(Y_n) \stackrel{(\text{Step 2})}{=} \lim_{n \rightarrow \infty} \mathbb{E}_{\mathbb{P}}(XY_n)$$

Since $X \geq 0$ and $Y_n \uparrow Y$, the sequence $(XY_n)_{n \in \mathbb{N}}$ monotonically increases to XY . Applying the Monotone Convergence Theorem under $\mathbb{P}$:

$$\lim_{n \rightarrow \infty} \mathbb{E}_{\mathbb{P}}(XY_n) = \mathbb{E}_{\mathbb{P}}(XY)$$

Hence, $\mathbb{E}_{\mathbb{Q}}(Y) = \mathbb{E}_{\mathbb{P}}(XY)$.

□

Part d)

We define the non-negative random variable $Y := \frac{1}{X} \mathbf{1}_{\{X>0\}}$ on the entire sample space Ω . Applying the formula proven in Part c) to the non-negative random variable Y , we obtain:

$$\mathbb{E}_{\mathbb{Q}}(Y) = \mathbb{E}_{\mathbb{P}}(XY) = \mathbb{E}_{\mathbb{P}}\left(X \cdot \frac{1}{X} \mathbf{1}_{\{X>0\}}\right)$$

For all $\omega \in \Omega$, the pointwise evaluation of $X(\omega)Y(\omega)$ yields 1 when $X(\omega) > 0$ and 0 when $X(\omega) = 0$. Thus, $XY = \mathbf{1}_{\{X>0\}}$ holds for all $\omega \in \Omega$. Then:

$$\mathbb{E}_{\mathbb{Q}}(Y) = \mathbb{E}_{\mathbb{P}}(\mathbf{1}_{\{X>0\}}) = \mathbb{P}(X > 0) = 1$$

Since $1 < \infty$, the expected value exists. Therefore, Y is integrable with respect to $\mathbb{Q}$. By assumption, $\mathbb{P}(X = 0) = 0$. By the property proven in Part b), this implies $\mathbb{Q}(X = 0) = 0$. Therefore, Y represents $\frac{1}{X}$ almost surely under both $\mathbb{P}$ and $\mathbb{Q}$. Since Y is integrable with respect to $\mathbb{Q}$ (i.e., $\mathbb{E}_{\mathbb{Q}}(Y) < \infty$) and $Y = \frac{1}{X}$ $\mathbb{Q}$ -almost surely, $\frac{1}{X}$ is integrable with respect to $\mathbb{Q}$. $\square$

Part e)

To evaluate $R(A) := \mathbb{E}_{\mathbb{Q}}\left(\frac{1}{X} \mathbf{1}_A\right)$, we interpret the expression inside the expectation as the strictly well-defined non-negative random variable $Y := \frac{1}{X} \mathbf{1}_{\{X>0\}} \mathbf{1}_A$ on Ω . Because $\mathbb{Q}(X = 0) = 0$ (from Part b), Y represents the definition of $\frac{1}{X} \mathbf{1}_A$ almost surely under $\mathbb{Q}$. Therefore, we evaluate $R(A) = \mathbb{E}_{\mathbb{Q}}(Y)$. Applying the formula from Part c) to Y , we obtain:

$$R(A) = \mathbb{E}_{\mathbb{Q}}(Y) = \mathbb{E}_{\mathbb{P}}(XY) = \mathbb{E}_{\mathbb{P}}\left(X \cdot \frac{1}{X} \mathbf{1}_{\{X>0\}} \mathbf{1}_A\right)$$

For all $\omega \in \Omega$, the pointwise evaluation of $X(\omega)Y(\omega)$ yields $\mathbf{1}_A(\omega)$ when $X(\omega) > 0$ and 0 when $X(\omega) = 0$. Thus, $XY = \mathbf{1}_{\{X>0\}} \mathbf{1}_A$ holds for all $\omega \in \Omega$. Since we are given that $\mathbb{P}(X > 0) = 1$, the random variable $\mathbf{1}_{\{X>0\}} \mathbf{1}_A$ is $\mathbb{P}$ -almost surely equivalent to $\mathbf{1}_A$. Because equivalent random variables possess the same expected value (Definition 3.3.8), we have:

$$R(A) = \mathbb{E}_{\mathbb{P}}(\mathbf{1}_{\{X>0\}} \mathbf{1}_A) = \mathbb{E}_{\mathbb{P}}(\mathbf{1}_A) = \mathbb{P}(A)$$

Since $R(A) = \mathbb{P}(A)$ holds for all $A \in \mathcal{F}$, we conclude that $R = \mathbb{P}$. $\square$